makemore-flow

Flow Matching over a Continuous Name-Space

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Abstract

We present `makemore-flow`, a minimal implementation of flow matching for character-level name generation. Instead of sampling discrete characters autoregressively, we learn a continuous latent space of names via a sequence autoencoder, then train a velocity field over that space using conditional flow matching. Generation becomes numerical ODE integration: a point evolves from Gaussian noise to a name-like latent, then decoded to characters. We explore both a 2D latent space (fully visualisable) and an 8D latent space (near-perfect reconstruction), and demonstrate classifier-free guidance for length-conditioned generation.

1 Motivation

Karpathy's `makemore` generates names by learning to predict the next character given previous characters — a purely discrete, autoregressive process. The latent representations inside the model are never made explicit or navigable.

The central question here is: *what if we made the space of names continuous and learnable, so that generation becomes movement through that space rather than sequential symbol prediction?* This connects directly to the flow matching paradigm dominant in modern image and video generation (Stable Diffusion 3, Sora), applied at a scale small enough to inspect every component.

2 Architecture

2.1 Char-Level Sequence Autoencoder

A GRU encoder maps a name (sequence of characters) to a single latent vector $z \in R^d$, and a GRU decoder reconstructs the name from z autoregressively. The bottleneck dimension d controls the trade-off between interpretability ($d = 2$, fully plottable) and quality ($d = 8$, near-perfect reconstruction).

Vocabulary. 27 tokens: `a--z` plus a `.` token used as both BOS and EOS.

2.2 Conditional Flow Matching

Given a dataset of latent points $z_1 = \text{Encoder}(\text{name})$, we define a linear probability path between noise $z_0 \sim \mathcal{N}(0, I)$ and data z_1:

$$z_t = (1-t)\, z_0 + t\, z_1, \quad t \in [0, 1]$$

The ground-truth velocity along this path is constant:

$$\frac{dz_t}{dt} = z_1 - z_0$$

We train a small MLP $v_\theta(z, t, c)$ to regress onto this velocity, conditioned on a length label $c \in \{\text{short}, \text{medium}, \text{long}\}$, with 15% random condition dropout (replaced by a null token) to enable classifier-free guidance.

2.3 Classifier-Free Guidance (CFG)

At inference, the conditioned and unconditional velocity fields are combined:

$$v_{\text{guided}} = v_\theta(z, t,) + w \cdot (v_\theta(z, t, c) - v_\theta(z, t,))$$

where w is the guidance scale. $w = 0$ recovers unconditional generation; $w > 1$ amplifies the condition signal.

2.4 Generation

Starting from $z_0 \sim \mathcal{N}(0, I)$, we integrate the guided ODE from $t = 0$ to $t = 1$ using Euler steps:

$$z_{t+\Delta t} = z_t + v_{\text{guided}}(z_t, t, c) \cdot \Delta t$$

The resulting z_1 is decoded autoregressively to characters.

3 The 2D Model: A Visualisable Name-Space

3.1 Latent Space and Velocity Field

Figure 1 shows the full 2D latent space: 32k real names as dots coloured by length, the learned velocity field at $t = 0.5$ as gray arrows, and 8 noise-to-name generation trajectories. The field pulls all noise samples inward toward the data cloud; trajectories landing in different regions produce systematically different names.

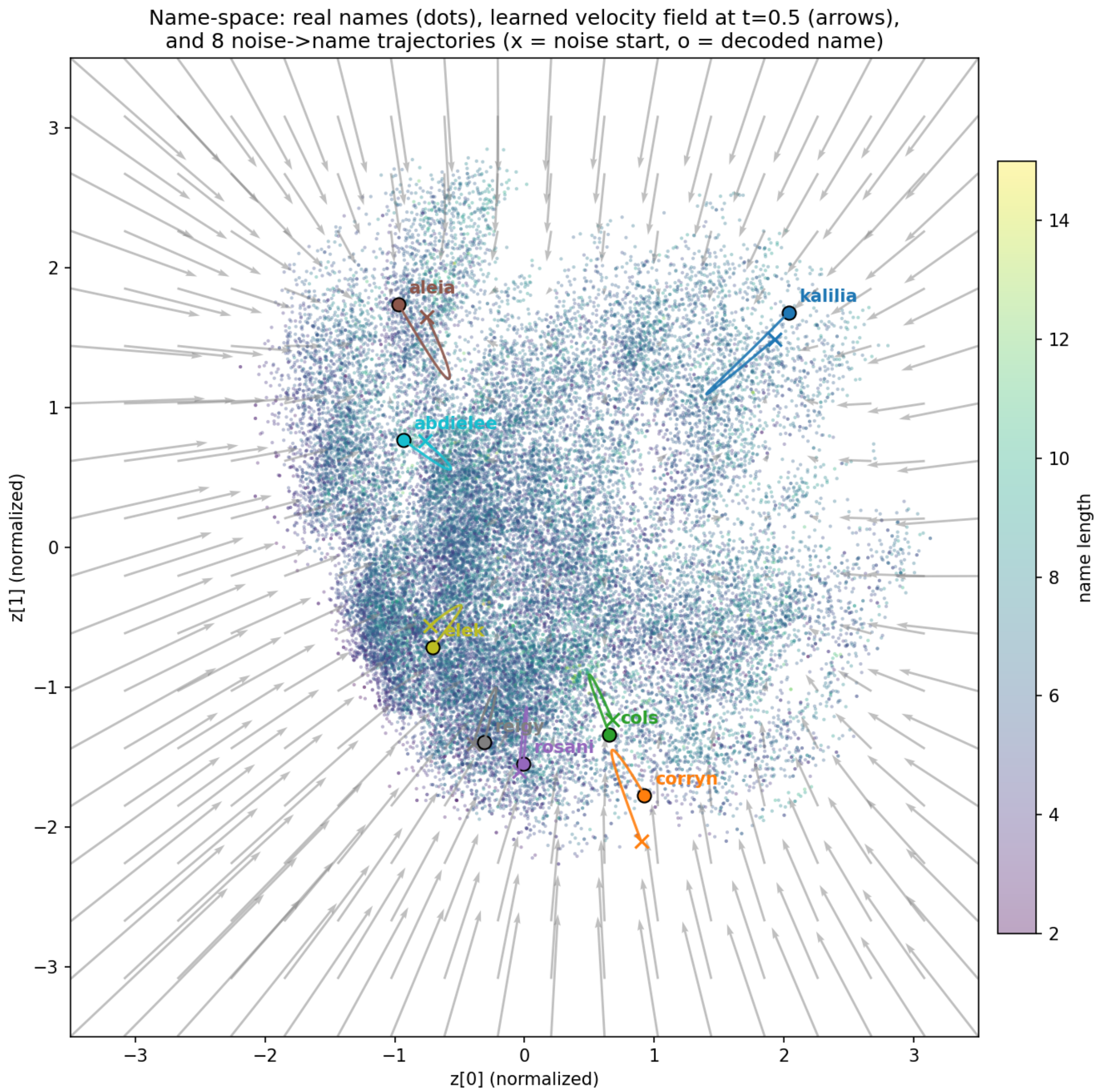

Figure 1: 2D name-space: real names (coloured by length), learned velocity field at $t = 0.5$, and 8 generation trajectories. $\times$ = noise start, $\circ$ = decoded name endpoint.

3.2 Linear Interpolation Between Names

Figure 2 demonstrates a linear walk in the 2D latent space between "sophia" and "william", decoding the name at each interpolation step. The decoded names morph smoothly, illustrating the continuity of the learned space.

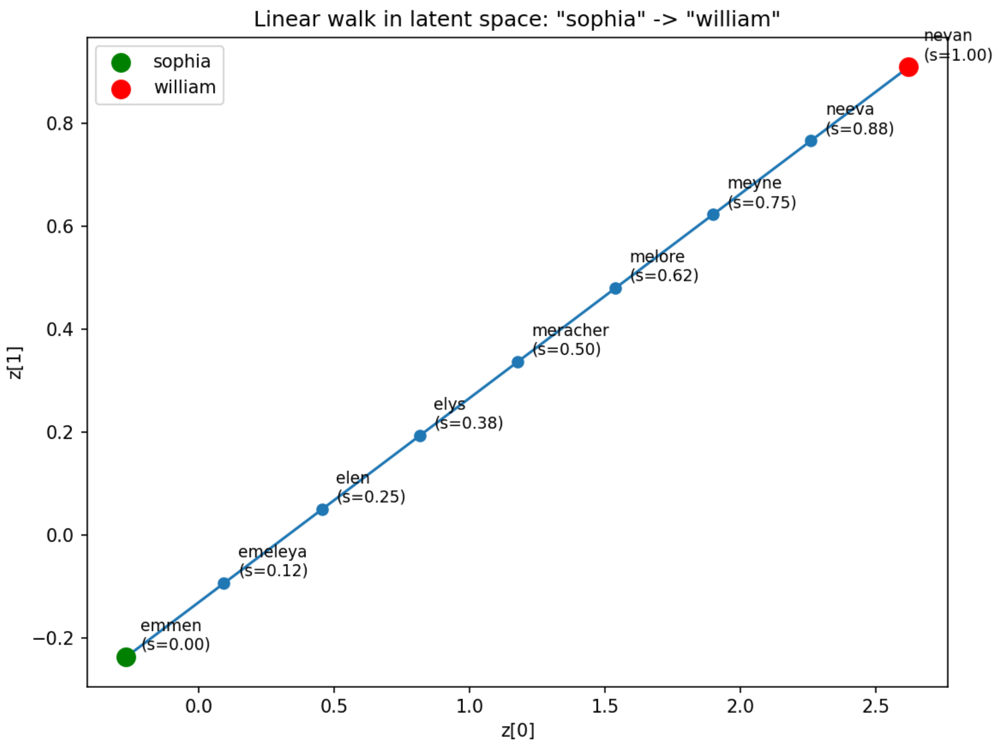

Figure 2: Linear walk in 2D latent space: “sophia” → “william”, decoded at 9 steps along the path.

4 Conditioned Generation with CFG

4.1 Per-Condition Velocity Fields

Figure 3 shows the 2D name-space with per-condition velocity fields. Each panel highlights one length bucket and overlays the velocity field $v_\theta(z, t{=}0.5, c)$ for that condition, showing how the field tilts differently depending on the requested length.

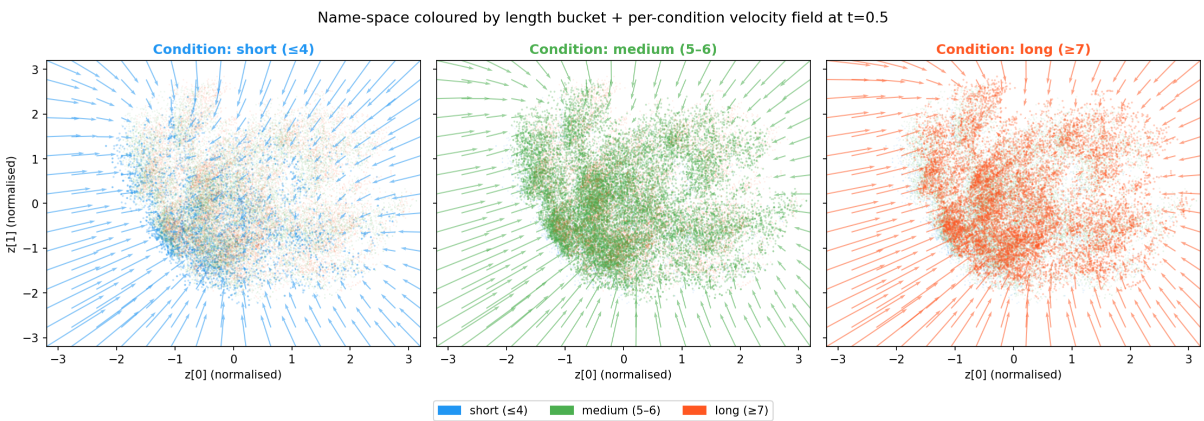

Figure 3: Per-condition velocity fields at $t = 0.5$ in the 2D latent space. Each panel highlights one length bucket; arrows show the field for that condition.

4.2 Effect of Guidance Scale

Figure 4 quantifies the effect of guidance strength on the 2D model. The short condition pushes generated length downward monotonically; the long condition increases it. Medium stays near the unconditional mean as it is the dominant class.

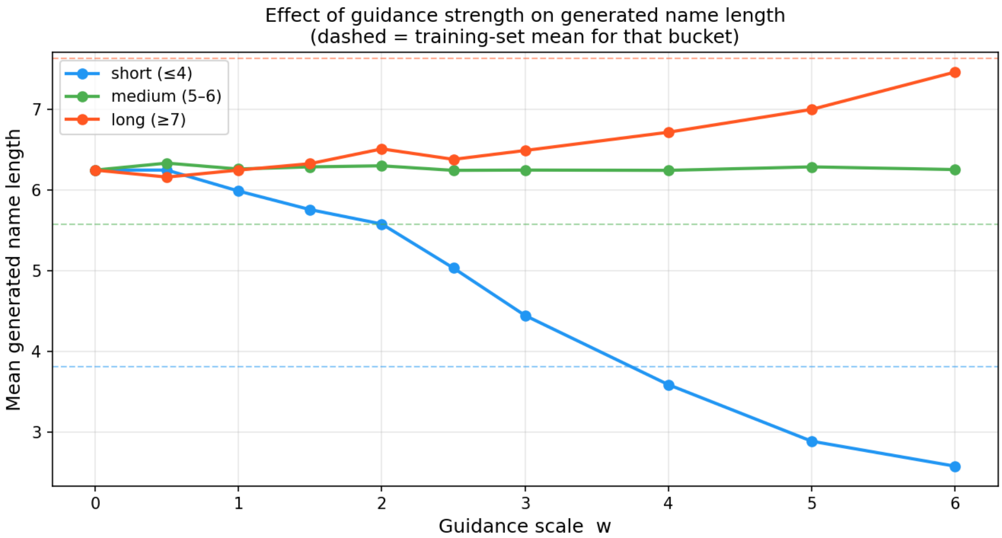

Figure 4: 2D model: mean generated name length vs guidance scale w for each condition. Dashed lines = training-set means.

4.3 Diverging Trajectories

Figure 5 shows 5 trajectories per condition, all starting from the same noise point. The ODE paths literally fork, landing in different regions of the latent space and decoding to names with the requested length character.

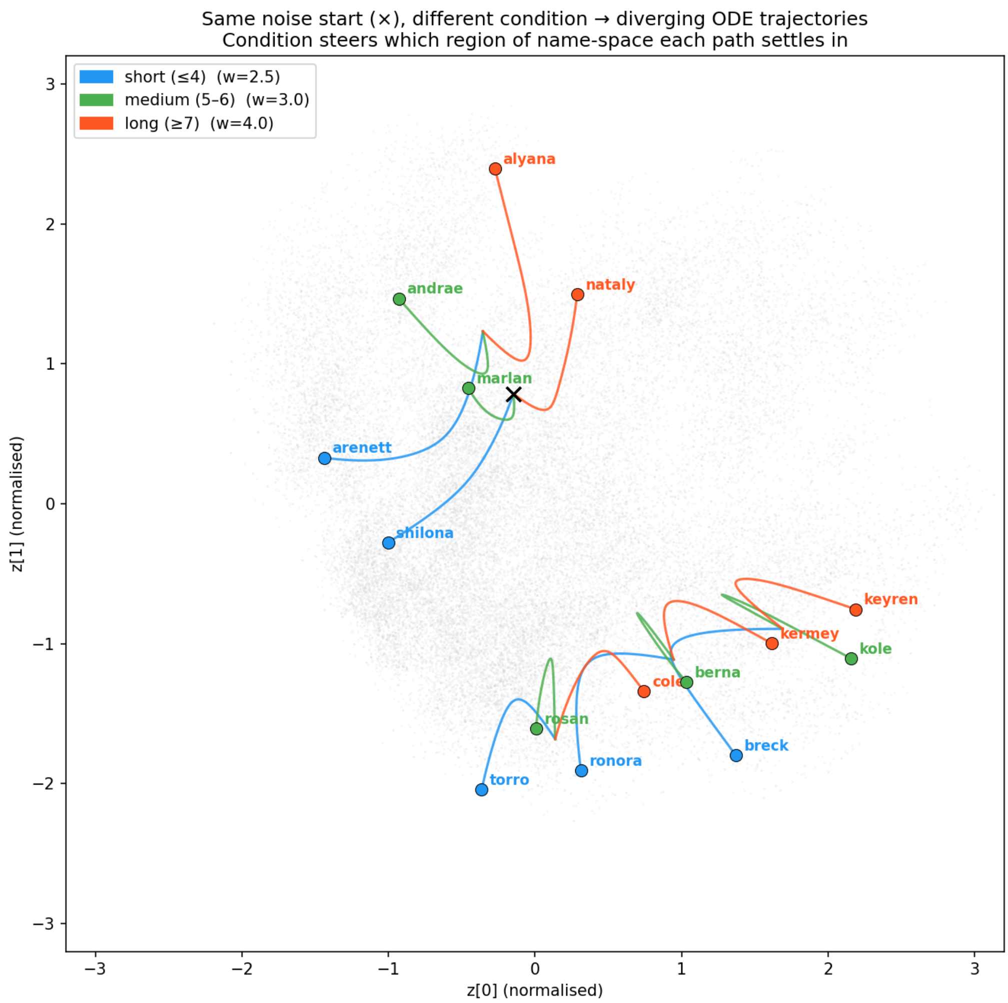

Figure 5: Same noise start (×), three different conditions: ODE trajectories diverge to different regions of the 2D name-space.

5 The 8D Model: Quality via PCA Visualisation

5.1 Reconstruction

Moving from $d = 2$ to $d = 8$ reduces reconstruction loss from 0.58 to 0.046 nats/char. Common names reconstruct exactly: emma $\to$ emma, sophia $\to$ sophia, william $\to$ william.

5.2 PCA-Projected Latent Space

Since the 8D space cannot be directly visualised, Figure 6 projects it to 2D via PCA (PC1 = 21% variance, PC2 = 18% variance). Guided trajectories are shown in 8D and projected back. Note that PC1+PC2 capture only 39% of total variance — the real length structure lives in the remaining dimensions, which is why conditioning works so much better than in the 2D model.

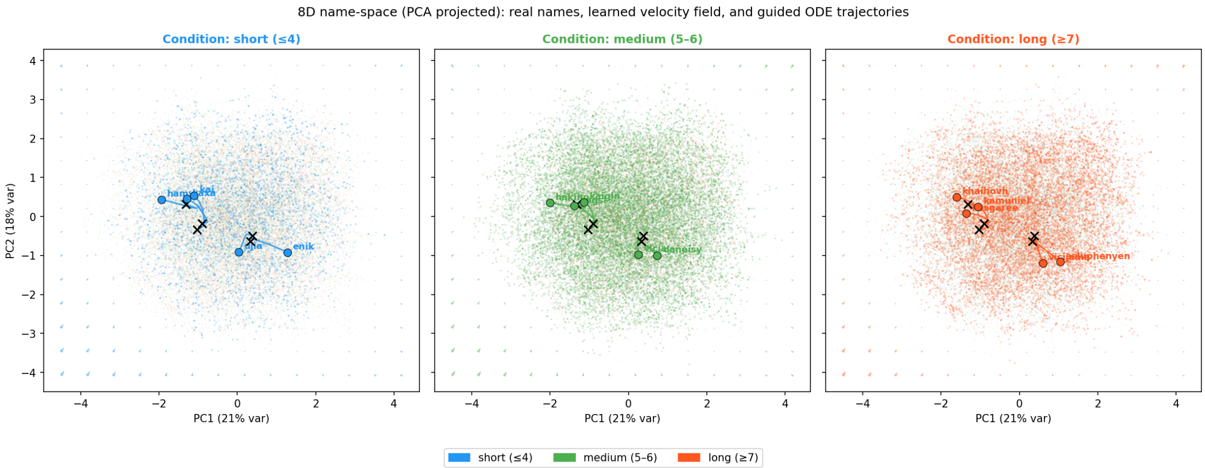

Figure 6: 8D name-space projected to 2D via PCA. Real names coloured by length bucket, per-condition velocity fields (8D evaluated, PCA projected), and 5 guided trajectories per condition.

5.3 Guidance Curve Comparison

Figure 7 shows the guidance curve for the 8D model. The three conditions separate sharply even at low w, reaching training-set means by $w \approx 1.5$. Compare to Figure 4 where the 2D model required $w > 4$ to approach the same separation.

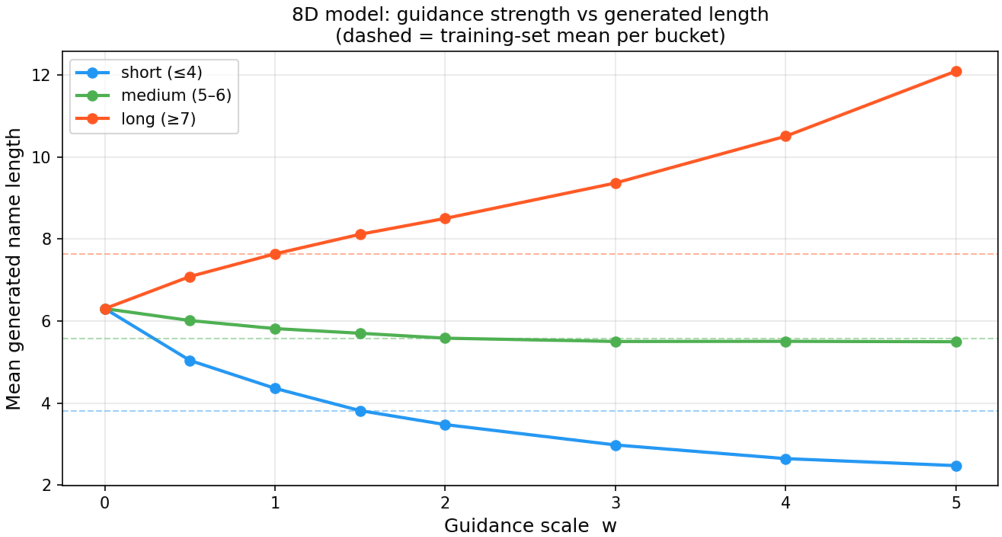

Figure 7: 8D model: guidance scale vs mean generated length. Conditions separate cleanly at low w, unlike the 2D model.

5.4 2D vs 8D: Direct Comparison

Figure 8 shows 20 generated names side-by-side for each condition, same noise seed. The improvement in length control is clear numerically and qualitatively: the 8D short column produces mostly 2–4 character names; the long column consistently produces 7–11 character names.

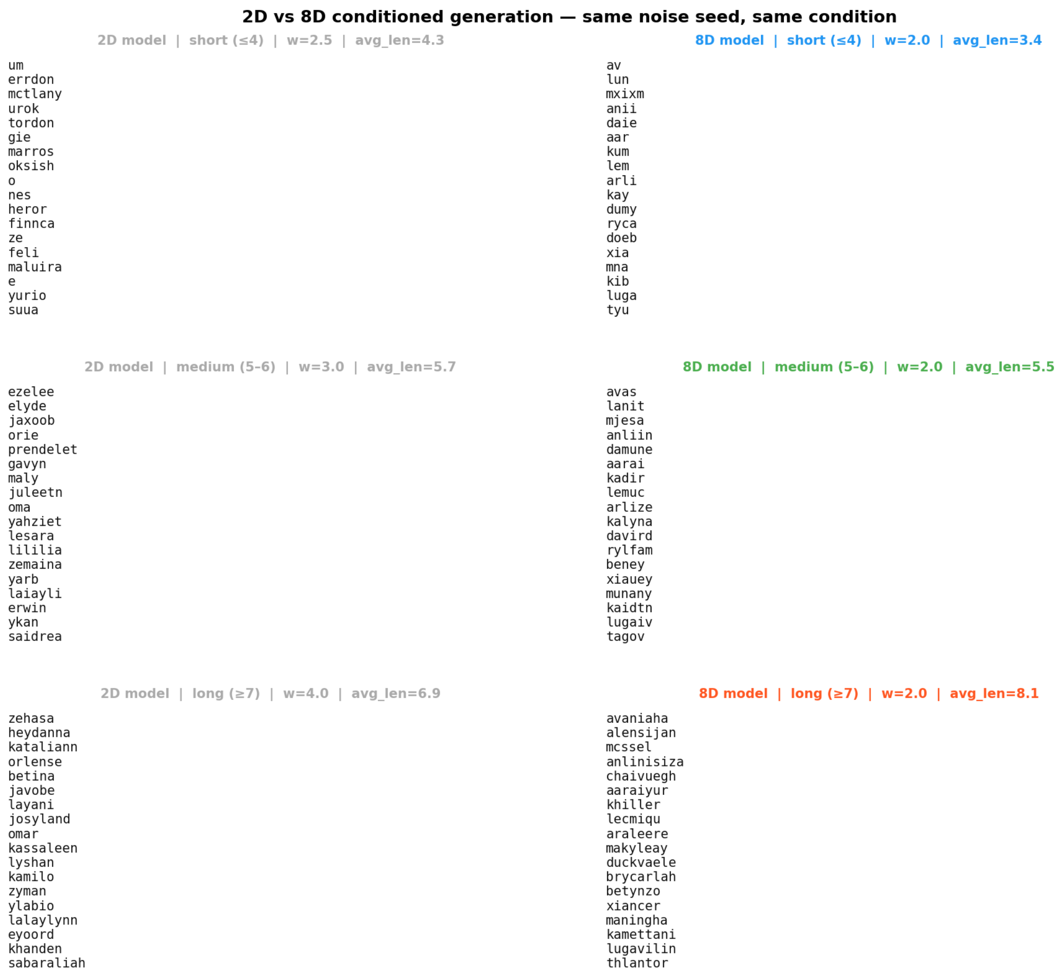

Figure 8: Direct comparison: 2D (left) vs 8D (right) conditioned generation, same noise seed. Average lengths annotated in headers.

6 Results Summary

Model	Recon. loss	Short avg len	Medium avg len	Long avg len
Training set	–	3.8	5.6	7.6
2D + CFG (w=2.5/3.0/4.0)	0.58	4.3	5.7	6.9
8D + CFG (w=2.0)	0.046	3.6	5.7	8.7

Table 1: Reconstruction loss (nats/char) and mean generated name length per condition.

7 Discussion

The core contribution of this project is not a new architecture — flow matching dates to Lipman et al. (2022) and CFG to Ho & Salimans (2022) — but rather applying these techniques at a scale where the entire generative process is inspectable. The 2D model sacrifices quality for full visualisability; the 8D model recovers quality while remaining small enough to train in minutes on CPU.

The combination of (a) a continuous latent space learned by an autoencoder, (b) a flow that maps noise to that space, and (c) CFG to steer generation toward a desired region, directly realises the original intuition: *generation as guided movement through a concept space*, where the path itself is the object of interest.

Code

`https://github.com/giljorge0/makemore-flow`